

Artificial Intelligence

ASSIGNMENT-II

UNIT-3

Q1. Explain reasoning under uncertainty in Artificial Intelligence. Why is uncertainty important in AI?

Answer: Reasoning under uncertainty is an important aspect of Artificial Intelligence because an intelligent agent may not always have complete or exact information about the environment. Uncertainty may arise because of **partial observability, non-determinism, incomplete information, or errors in the available data**. In such situations, the agent may not know the exact current state or the exact outcome of an action.

In ordinary logical reasoning, knowledge is often represented using rules such as:

$$A \rightarrow B$$

which means that if A is true, then B is true. But in many real-world situations, the system may not know whether A is completely true. Therefore, simple true-or-false logic is not always sufficient, and the AI system has to reason with uncertain knowledge.

A common example is dental diagnosis. Consider the rule:

Toothache $\rightarrow$ Cavity

This rule is not always correct because a patient with a toothache may have a cavity, but the cause may also be gum disease, an abscess, or some other problem. Hence, the AI system should not treat the conclusion as absolutely certain. Instead, it should assign a **degree of belief** to different possible causes.

The major causes of uncertainty include:

1. Information obtained from unreliable sources
2. Experimental errors
3. Equipment faults
4. Temperature variations
5. Climate changes

To handle uncertainty, AI makes use of **probability theory**. Probability provides a numerical way to represent the likelihood of an event. The probability of an event always lies between 0 and 1.

$$0 \leq P(A) \leq 1$$

Here, **P(A) = 0** indicates that the event is impossible, while **P(A) = 1** indicates complete certainty. Values between 0 and 1 represent different degrees of uncertainty.

Probabilistic reasoning is a method of knowledge representation in which probability is used to indicate uncertainty. It combines **probability theory with logic** so that an AI system can reason when knowledge is incomplete or uncertain. Examples include statements such as “It will rain today,” “a person may behave in a particular way,” or “one team may win a match.” Such statements are not completely certain and therefore require probabilistic reasoning.

Probabilistic reasoning is needed in AI in situations such as:

- when outcomes are unpredictable,
- when the number of possibilities becomes too large to handle directly,
- and when unknown errors occur during an experiment.

Two important methods used for handling uncertain knowledge are:

1. Bayes' Rule

2. Bayesian Statistics

Bayes' Rule is particularly useful because it allows an AI system to update the probability of an event when new evidence becomes available.

Uncertainty is also important in rational decision-making. An AI agent may have several possible actions, and each action may lead to different outcomes. To choose among them, the agent needs to consider both the **probability of an outcome** and its **utility or usefulness**. The relation is expressed as:

Decision Theory = Probability Theory + Utility Theory

A rational agent selects the action that produces the **highest expected utility** over all possible outcomes. This is called the **Maximum Expected Utility (MEU) principle**.

Importance of Uncertainty in AI

Uncertainty is important because real-world environments are rarely completely predictable. AI systems often work with incomplete observations, noisy information, changing conditions, and uncertain outcomes. Therefore, they must be able to make decisions or predictions even when complete knowledge is not available.

Without uncertainty handling, an AI system would be useful only in perfectly known situations. By using probability and decision theory, AI systems can behave more realistically and make better decisions in practical applications such as medical diagnosis, weather prediction, robotics, finance, and expert systems.

Q2. Explain Bayesian probabilistic inference. Describe how prior probability is updated using new evidence.

Answer: Bayesian probabilistic inference is a method used in Artificial Intelligence to reason with uncertain information. It is based on Bayes' theorem, also called Bayes' rule or Bayesian reasoning. It determines the probability of an event when new evidence becomes available and provides a systematic way to revise an existing belief in the light of new information.

In Bayesian inference, an AI system starts with an initial belief about a hypothesis or event. This initial belief is called the **prior probability**. When new evidence is observed, the system combines the prior probability with the likelihood of that evidence. The result is a revised probability called the **posterior probability**.

Bayes' Theorem

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

where:

- $P(A)$ = Prior probability of event A.
- $P(B|A)$ = Likelihood, i.e., probability of observing evidence B when A is true.
- $P(B)$ = Marginal probability or probability of the evidence.
- $P(A|B)$ = Posterior probability of A after observing evidence B.

Thus, Bayesian inference can be expressed as:

$$\text{Posterior Probability} = \frac{\text{Likelihood} \times \text{Prior Probability}}{\text{Evidence}}$$

Derivation of Bayes' Theorem

From the definition of conditional probability:

$$P(A \cap B) = P(A | B) P(B)$$

Similarly,

$$P(A \cap B) = P(B | A) P(A)$$

Since both expressions represent the same joint probability,

$$P(A | B) P(B) = P(B | A) P(A)$$

Therefore,

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Updating Prior Probability Using New Evidence

The Bayesian updating process takes place in the following steps:

1. Start with a prior probability: This represents the initial belief about an event before considering new evidence.
2. Observe new evidence: New information is obtained from the environment or from an experiment.
3. Calculate the likelihood: Determine how likely the observed evidence is if the hypothesis is true.
4. Apply Bayes' theorem: Combine the prior probability and likelihood and normalize by the probability of the evidence.
5. Obtain the posterior probability: The resulting posterior represents the updated belief after considering the evidence.

Prior Probability + New Evidence → Bayes' Theorem → Posterior Probability

The important idea is that the posterior probability obtained after one observation can be used as the prior probability when further evidence becomes available. Thus, an AI system can continuously revise its beliefs as it receives new information.

Example: Meningitis and Stiff Neck

Suppose a doctor wants to determine the probability that a patient has meningitis given that the patient has a stiff neck.

Let M = patient has meningitis and S = patient has a stiff neck. The given probabilities are:

$$P(S | M) = 0.8$$

$$P(M) = \frac{1}{30000}$$

$$P(S) = 0.02$$

Using Bayes' theorem:

$$P(M | S) = \frac{P(S | M) P(M)}{P(S)}$$

Substituting the values:

$$P(M | S) = \frac{0.8 \times \frac{1}{30000}}{0.02}$$

$$P(M | S) = \frac{1}{750}$$

Therefore, after observing the evidence of a stiff neck, the probability that the patient has meningitis is approximately 1 in 750. This example shows how a prior belief is updated when new evidence is observed.

Importance of Bayesian Probabilistic Inference in AI

- It represents uncertain knowledge numerically using probabilities.
- It combines existing knowledge with newly observed evidence.
- It allows predictions and beliefs to be updated dynamically.
- It supports rational decision-making when complete information is unavailable.
- It is useful in diagnosis, prediction, classification and decision-support systems.

Q3. What is Dempster–Shafer Theory? Explain its basic concepts of belief, plausibility, and evidence.

Answer:

Dempster–Shafer Theory (DST), also known as the Theory of Evidence, is a mathematical framework for reasoning under uncertainty. It is useful when the available information is incomplete, uncertain, or obtained from different sources. Unlike ordinary probability, DST can assign support to a single hypothesis as well as to a set of possible hypotheses.

1. Frame of Discernment

The frame of discernment is the set of all possible hypotheses or outcomes under consideration. It is usually represented by Θ (theta).

Example: $\Theta = \{D_1, D_2, D_3\}$,

where D_1 , D_2 and D_3 are three possible diseases.

2. Evidence and Basic Probability Assignment

Evidence is the information obtained from sources such as experts, sensors, observations, experiments, or test results. In Dempster–Shafer Theory, the strength of evidence is represented by a Basic Probability Assignment (BPA), also called a mass function, denoted by $m(A)$.

The mass function satisfies: $m(\emptyset) = 0$ and $\sum m(A) = 1$.

An important feature of DST is that evidence may be assigned to a set of hypotheses when the evidence does not identify one hypothesis exactly.

Example: $m(\{D_1\}) = 0.6$, $m(\{D_1, D_2\}) = 0.3$, and $m(\Theta) = 0.1$. Here, 0.6 directly supports D_1 , 0.3 supports either D_1 or D_2 , and 0.1 represents remaining uncertainty.

3. Belief

Belief, denoted by $\text{Bel}(A)$, represents the total amount of evidence that definitely supports a proposition A . It is the minimum degree of confidence justified by the available evidence.

$$\text{Bel}(A) = \sum m(B), \text{ for all } B \subseteq A$$

The value of belief lies between 0 and 1: $0 \leq \text{Bel}(A) \leq 1$.

For example, if the direct evidence supporting disease D_1 is 0.6, then $\text{Bel}(D_1) = 0.6$. This means that there is at least 60% confirmed support for D_1 .

4. Plausibility

Plausibility, denoted by $\text{Pl}(A)$, represents the maximum possible support that may be given to a proposition A . It includes all evidence that does not contradict A .

$$\text{Pl}(A) = \sum m(B), \text{ for all } B \text{ such that } B \cap A \neq \emptyset$$

It can also be written as: $\text{Pl}(A) = 1 - \text{Bel}(\bar{A})$

Belief is the minimum confirmed support, whereas plausibility is the maximum possible support. Therefore:

$$\text{Bel}(A) \leq \text{Pl}(A)$$

The interval $[\text{Bel}(A), \text{Pl}(A)]$ represents the uncertainty associated with proposition A . The difference $\text{Pl}(A) - \text{Bel}(A)$ indicates the amount of remaining uncertainty.

5. Example of Belief and Plausibility

Suppose an AI medical system evaluates a disease D and obtains $\text{Bel}(D) = 0.65$ and $\text{Pl}(D) = 0.85$. This means that the available evidence gives at least 65% definite support to D , while as much as 85% support is possible when uncertain evidence is also considered.

Thus, the uncertainty interval is:

$$[\text{Bel}(D), \text{Pl}(D)] = [0.65, 0.85]$$

6. Combining Evidence

One of the important features of Dempster-Shafer Theory is the ability to combine evidence obtained from different independent sources. Dempster's Rule of Combination is used to combine these pieces of evidence and produce an updated assessment. For example, in medical diagnosis, evidence may come from symptoms, laboratory tests, and expert observations.

Importance in Artificial Intelligence

Dempster-Shafer Theory is useful in AI because intelligent systems often have to work with incomplete or uncertain information. It allows the system to distinguish between confirmed support and possible support, represent ignorance explicitly, and combine evidence from multiple sources. It is useful in expert systems, diagnosis, sensor fusion, pattern recognition, and decision-support systems.

Unit 4

Q1. Explain First Order Logic (FOL) with its syntax, semantics, and examples.

Answer: First Order Logic (FOL), also known as predicate logic or first-order predicate calculus, is a formal system used to represent knowledge about objects, their properties, and the relationships among objects. In Artificial Intelligence, FOL is important for knowledge representation, automated reasoning, and natural language processing.

FOL is more expressive than propositional logic because it can describe individual objects, properties, relationships, and quantified statements.

For example, the statement “Socrates is a man” can be represented as:

Man(Socrates)

The general statement “All men are mortal” can be represented as:

$\forall x [Man(x) \rightarrow Mortal(x)]$

1. Syntax of First Order Logic

The syntax of FOL specifies the symbols and rules used to construct valid logical expressions. The main components are constants, variables, predicates, functions, quantifiers, and logical connectives.

1.1 Constants

Constants represent specific objects in the domain of discourse. Examples include Alice, Bob, John, 2, and NewYork.

Student(John)

1.2 Variables

Variables represent unspecified objects in the domain. Common variables are x, y, and z.

Student(x)

1.3 Predicates

Predicates represent properties of objects or relationships between objects and evaluate to true or false.

Student(Ravi)

Likes(Alice, Bob)

GreaterThan(x, 2)

1.4 Functions

Functions map one or more objects to another object. For example, $MotherOf(x)$ denotes the mother of x .

$$MotherOf(x)$$

1.5 Quantifiers

The two important quantifiers in FOL are the universal and existential quantifiers.

Universal Quantifier ($\forall$): It means “for all” or “for every.”

$$\forall x [Person(x) \rightarrow Mortal(x)]$$

This means: All persons are mortal.

Existential Quantifier ($\exists$): It means “there exists at least one.”

$$\exists x [Person(x) \wedge Likes(x, IceCream)]$$

This means: There exists a person who likes ice cream.

1.6 Logical Connectives

Symbol	Name	Meaning
$\neg$	Negation	NOT
$\wedge$	Conjunction	AND
$\vee$	Disjunction	OR
$\rightarrow$	Implication	IF...THEN
$\leftrightarrow$	Biconditional	IF AND ONLY IF

Example:

$$Student(Ravi) \wedge Intelligent(Ravi)$$

2. Structure of an FOL Expression

A First Order Logic expression is built from terms and predicates. Terms may be constants, variables, or functions. Predicates describe properties or relationships. Quantifiers and logical connectives are then used to form complete sentences.

FIRST ORDER LOGIC		
TERMS	PREDICATES	SENTENCES
Constants Variables Functions	Properties Relationships	Predicates + Quantifiers + Connectives

Basic structure of First Order Logic

For the expression:

$$\forall x [Student(x) \rightarrow Intelligent(x)]$$

$\forall$ is the universal quantifier, x is a variable, Student and Intelligent are predicates, and $\rightarrow$ is the implication connective.

3. Semantics of First Order Logic

Syntax tells us how to construct a valid FOL expression, whereas semantics tells us what the expression means and whether it is true or false under a particular interpretation.

The semantics of FOL is based on an interpretation of a domain. A domain is the collection of objects being discussed. Constants identify particular objects, predicates identify properties or relations, and functions identify mappings between objects.

$$D = \{Ravi, Sita, John\}$$

For example, Student(Ravi) is true if Ravi belongs to the set of students in the chosen interpretation. Similarly, Likes(Ravi, AI) is true if the relationship Likes holds between Ravi and AI.

4. Example of FOL Representation and Reasoning

Consider the following statements:

1. All humans are mortal.
2. Socrates is a human.
3. Therefore, Socrates is mortal.

The FOL representation is:

$$\forall x [Human(x) \rightarrow Mortal(x)]$$

$$Human(Socrates)$$

Therefore, we can infer:

$$Mortal(Socrates)$$

5. Knowledge Base Example

Consider the following knowledge base:

$$\forall x [Cat(x) \rightarrow Mammal(x)]$$

$$\forall x [Mammal(x) \rightarrow Animal(x)]$$

$$Cat(Tom)$$

From these statements, we infer:

$$Mammal(Tom)$$

$$Animal(Tom)$$

6. More Examples of English Sentences in FOL

All students are intelligent.

$$\forall x [\textit{Student}(x) \rightarrow \textit{Intelligent}(x)]$$

Some students like AI.

$$\exists x [\textit{Student}(x) \wedge \textit{Likes}(x, \textit{AI})]$$

Every person likes someone.

$$\forall x [\textit{Person}(x) \rightarrow \exists y \textit{Likes}(x, y)]$$

Marcus was a man.

$$\textit{Man}(\textit{Marcus})$$

Marcus was a Pompeian.

$$\textit{Pompeian}(\textit{Marcus})$$

All Pompeians were Romans.

$$\forall x [\textit{Pompeian}(x) \rightarrow \textit{Roman}(x)]$$

6. Propositional Logic vs First Order Logic

7.

Feature	Propositional Logic	First Order Logic
Basic Unit	Propositions	Predicates, constants, variables
Expressiveness	Limited true/false statements	Objects, properties and relationships
Quantifiers	Not available	Universal ($\forall$) and Existential ($\exists$)
Semantics	Truth tables	Interpretation of a domain
Example	$P \rightarrow Q$	$\forall x [\textit{Human}(x) \rightarrow \textit{Mortal}(x)]$

8. Applications of FOL in Artificial Intelligence

- Knowledge Representation
- Automated Theorem Proving
- Natural Language Processing (NLP)
- Expert Systems
- Semantic Web

Q2. Explain Inference in First Order Logic. Discuss Unification and Resolution.

Answer: Inference in First Order Logic (FOL) is the process of deriving new facts or sentences from existing facts and rules in a knowledge base. It enables an AI system to use stored knowledge to reach conclusions that are not explicitly stated.

For example, suppose the knowledge base contains:

$$\forall x [\mathbf{Human}(x) \rightarrow \mathbf{Mortal}(x)]$$

and

$$\mathbf{Human}(\mathbf{Socrates})$$

From these statements, we can infer:

$$\mathbf{Mortal}(\mathbf{Socrates})$$

$$\mathbf{Facts} + \mathbf{Rules} \rightarrow \mathbf{Inference} \rightarrow \mathbf{New\ Facts}$$

1. Substitution in FOL

Substitution is the process of replacing a variable with a constant or another term. If $F[a/x]$ is written, it means that the constant a is substituted for the variable x .

$$F[a/x]$$

Example:

$$\forall x [\mathbf{King}(x) \wedge \mathbf{Greedy}(x) \rightarrow \mathbf{Evil}(x)]$$

Substituting John for x gives:

$$\mathbf{King}(\mathbf{John}) \wedge \mathbf{Greedy}(\mathbf{John}) \rightarrow \mathbf{Evil}(\mathbf{John})$$

Substitution is fundamental to inference because FOL rules commonly contain variables.

2. Unification

Unification is the process of finding a substitution that makes two different logical expressions identical. It accepts two literals or expressions and determines suitable substitutions for their variables so that the expressions match.

2.1 Simple Example

Consider:

$$\mathbf{Knows}(\mathbf{John}, x)$$

and

$$\mathbf{Knows}(\mathbf{John}, \mathbf{Mary})$$

To make them identical, substitute:

$$x = \mathbf{Mary}$$

Hence, the unifier is:

$$\{x/\mathbf{Mary}\}$$

2.2 Another Example

Consider the two expressions:

$$\begin{array}{l} P(x, y) \\ P(a, f(z)) \end{array}$$

They can be unified by the substitutions:

$$x = a, y = f(z)$$

Thus:

$$\{x/a, y/f(z)\}$$

After substitution, both expressions become:

$$P(a, f(z))$$

2.3 Most General Unifier (MGU)

A substitution that makes two expressions identical without unnecessary substitutions is called a Most General Unifier (MGU).

$$\begin{array}{l} P(x, John) \\ P(Mary, y) \end{array}$$

An MGU is:

$$\{x/Mary, y/John\}$$

2.4 Failure of Unification

Unification fails when expressions contain incompatible constants or structures. For example:

$$\begin{array}{l} Likes(John, Mango) \\ Likes(John, Apple) \end{array}$$

These cannot be unified by variable substitution because Mango and Apple are different constants.

3. Resolution

Resolution is an inference rule that operates on expressions represented in Conjunctive Normal Form (CNF) or clausal form. It is commonly used to prove a statement by contradiction.

The basic idea is to negate the statement to be proved, add the negation to the knowledge base, and apply resolution. If a contradiction is produced, the original statement is proved.

3.1 Basic Resolution Example

Consider the clauses:

$$\begin{array}{l} P \vee Q \\ \neg Q \vee R \end{array}$$

The complementary literals Q and $\neg Q$ are eliminated, producing the resolvent:

$$P \vee R$$

The resolution rule can be written as:

$$(P \vee Q), (\neg Q \vee R) \Rightarrow (P \vee R)$$

4. Resolution in First Order Logic

In FOL, resolution also uses unification because predicates may contain variables.
Consider:

$$\neg \text{Human}(x) \vee \text{Mortal}(x) \\ \text{Human}(\text{Socrates})$$

First unify the variable:

$$x = \text{Socrates}$$

The first clause becomes:

$$\neg \text{Human}(\text{Socrates}) \vee \text{Mortal}(\text{Socrates})$$

Resolving it with Human(Socrates) gives:

$$\text{Mortal}(\text{Socrates})$$

Thus, unification finds matching literals and resolution uses those literals to derive a new conclusion.

5. Steps in Resolution

- Convert the given facts and rules into First Order Logic.
- Convert the FOL statements into Conjunctive Normal Form (CNF). This may involve eliminating implications, moving negations inward, standardizing variables, handling quantifiers, and producing clauses.
- Negate the statement that has to be proved.
- Apply unification to identify matching predicates or literals.
- Apply the resolution rule repeatedly to complementary literals.
- If an empty clause (contradiction) is obtained, the original goal is proved.

The empty clause is commonly represented as:

□

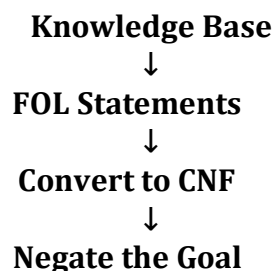
6. Relationship Between Unification and Resolution

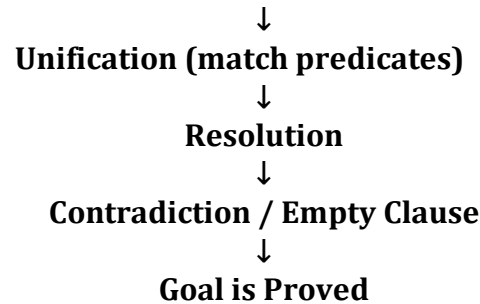
Unification	Resolution
Finds suitable substitutions.	Derives new clauses.
Makes expressions match.	Eliminates complementary literals.
Works with variables and terms.	Works with clauses.
Supports the inference process.	Acts as an inference rule.
Example: $x = \text{John}$	Example: resolve P with $\neg P$

A simple way to remember the difference is:

$$\text{Unification} = \text{Matching} \\ \text{Resolution} = \text{Reasoning}$$

8. Overall Inference Process





Q3. Compare Propositional Inference with First Order Logic Inference

Answer: Inference is the process of deriving new facts or conclusions from existing knowledge. Both Propositional Logic (PL) and First Order Logic (FOL) support inference, but they differ in representation power and reasoning mechanisms. Propositional logic deals mainly with complete true/false statements, whereas FOL represents objects, properties, relationships, variables and quantifiers.

1. Propositional Inference

In propositional logic, knowledge is represented using propositions such as P, Q and R. Each proposition is a complete statement that can be true or false.

$$P \rightarrow Q$$

If P is true, Modus Ponens gives:

$$P, \quad P \rightarrow Q \Rightarrow Q$$

Propositional inference works with complete propositions and does not represent the internal structure of objects or relationships.

2. First Order Logic Inference

FOL inference derives new facts from sentences containing predicates, constants, variables, functions and quantifiers. It can apply a general rule to individual objects.

$$\forall x [Human(x) \rightarrow Mortal(x)]$$

$$Human(Socrates)$$

Substituting Socrates for x gives:

$$Mortal(Socrates)$$

3. Variables and Quantifiers

Propositional logic does not use variables or quantifiers. FOL uses universal and existential quantifiers.

$$\forall = \text{Universal Quantifier (for all)}$$

$$\exists = \text{Existential Quantifier (there exists)}$$

$$\forall x [Student(x) \rightarrow Intelligent(x)]$$

4. Substitution in FOL

Because FOL contains variables, substitution is an important operation.

$$\forall x [King(x) \wedge Greedy(x) \rightarrow Evil(x)]$$

Substituting John for x:

$$King(John) \wedge Greedy(John) \rightarrow Evil(John)$$

5. Unification in FOL

Unification finds a substitution that makes two logical atomic expressions identical.

$$\begin{aligned} & Likes(x, AI) \\ & Likes(Ravi, AI) \end{aligned}$$

Unifier:

$$\{x/Ravi\}$$

Propositional inference does not require this type of unification because propositions do not contain variables or predicate arguments.

6. Comparison

Feature	Propositional Inference	First Order Logic Inference
Basic unit	Propositions	Predicates, constants and variables
Representation	Simple true/false statements	Objects, properties and relationships
Variables	Not used	Used
Quantifiers	Not available	Uses $\forall$ and $\exists$
Substitution	Generally not required	Important
Unification	Not required	Important for matching predicates
Expressiveness	Limited	More expressive
Semantics	Truth values / truth tables	Interpretation of a domain
Example	$P \rightarrow Q$	$\forall x [Human(x) \rightarrow Mortal(x)]$
Suitable for	Relatively simple logical problems	Complex knowledge and relationships

7. Example Showing the Difference

Statement: All students are intelligent, and Ravi is a student. Therefore, Ravi is intelligent.

Propositional Logic

$$\begin{aligned} P &= \text{Ravi is a student} \\ Q &= \text{Ravi is intelligent} \\ P &\rightarrow Q \\ P, P &\rightarrow Q \Rightarrow Q \end{aligned}$$

First Order Logic

$$\begin{aligned} \forall x [& \textit{Student}(x) \rightarrow \textit{Intelligent}(x)] \\ & \textit{Student}(\textit{Ravi}) \\ & x = \textit{Ravi} \\ & \textit{Intelligent}(\textit{Ravi}) \end{aligned}$$

The same general FOL rule can be applied to other students without writing a separate rule for each person.

Q4. Explain Decision Tree Learning. Discuss its Construction and Advantages.

Answer: Decision Tree Learning is a supervised learning technique used for both classification and regression, although it is mainly preferred for classification problems. It is a tree-structured classifier in which internal nodes represent features or attributes, branches represent decision rules, and leaf nodes represent the final outcome or class.

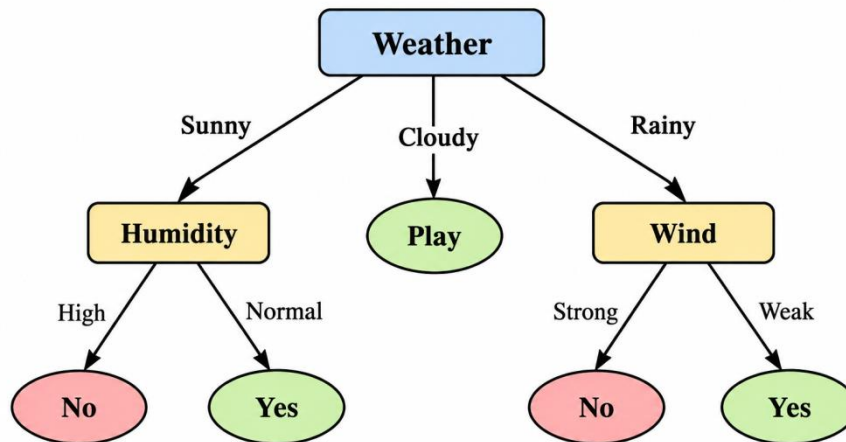
Decision tree learning is related to inductive learning, where a general rule is learned from a collection of input-output examples and is then used to predict the output for new examples.

Structure of a Decision Tree

Root Node: The topmost node of the tree. It represents the first attribute on which the dataset is divided.

Decision/Internal Node: Represents a test on an attribute. Depending on the result, a particular branch is followed.

Leaf/Terminal Node: Represents the final class or predicted output.



2. Construction of a Decision Tree

A decision tree is constructed from the training examples by repeatedly selecting an attribute and dividing the examples according to that attribute.

Step 1: Start with the Training Dataset

The algorithm starts with a set of labeled training examples. For illustration:

Weather	Humidity	Wind	Play
Sunny	High	Weak	No
Sunny	Normal	Weak	Yes
Cloudy	High	Strong	Yes
Rainy	High	Strong	No
Rainy	Normal	Weak	Yes

Step 2: Select the Best Attribute

A suitable attribute is selected to divide the examples. A good split separates examples into groups that are as pure as possible. Common measures include entropy and information gain.

Entropy measures the uncertainty or impurity in a dataset:

$$\text{Entropy}(S) = -\sum_i p_i \log_2(p_i)$$

For binary classification:

$$\text{Entropy}(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

Step 3: Calculate Information Gain

Information Gain measures the reduction in uncertainty after splitting the dataset using an attribute.

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum_v (|S_v|/|S|) \text{Entropy}(S_v)$$

An attribute with a high information gain provides a useful split and may be selected as the decision node.

Step 4: Create the Root Node

The selected attribute becomes the root node. For example, if Weather is selected, the first split may be Sunny, Cloudy and Rainy.

Step 5: Divide the Dataset

The training examples are divided into subsets according to the possible values of the selected attribute.

Step 6: Repeat the Process

For each subset, another suitable attribute is selected and a new decision node is created. This process is repeated recursively.

Step 7: Create Leaf Nodes

A leaf node is created when the examples in a subset belong to the same class, when no useful attributes remain, or when another stopping condition is satisfied.

Leaf Node → Final Prediction / Class

3. Working Example

Suppose a new example has Weather = Sunny and Humidity = Normal. Starting from the root, the decision path is:

Weather → Sunny → Humidity → Normal → Yes

Therefore, the predicted class is:

Play = Yes

4. Advantages of Decision Tree Learning

1. **Easy to understand:** The tree structure is simple and can be understood by students and users.
2. **Easy to interpret:** The path from the root to a leaf shows how a prediction is obtained.
3. **Classification and regression:** Decision trees can be used for both supervised classification and regression problems.
4. **Clear decision rules:** A root-to-leaf path can be represented as IF-THEN rules.
5. **Handles multiple attributes:** The method can learn decisions involving several input features.
6. **Useful for prediction:** A learned tree can be used to predict the class or output of unseen examples.
7. **Visual representation:** The graphical tree representation makes the decision-making process easy to explain.

Example decision rule:

IF Weather = Sunny AND Humidity = Normal THEN Play = Yes

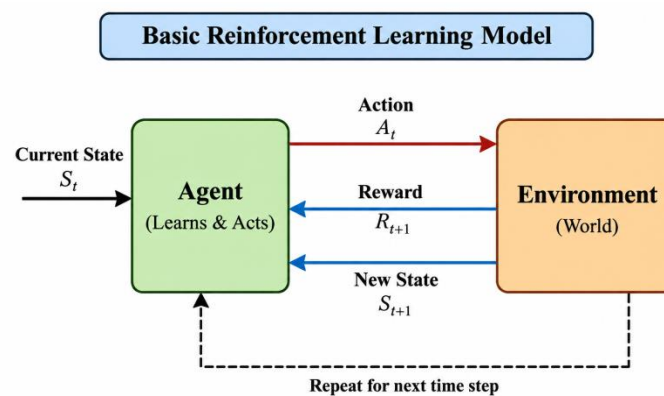
Q5. Explain Reinforcement Learning. Discuss Agent, Environment, Reward, and Policy.

Answer: Reinforcement Learning (RL) is a branch of Machine Learning in which an agent learns how to make decisions by interacting with an environment. The agent performs actions, observes the results, and receives feedback in the form of rewards or penalties. The main objective is to learn a behavior that maximizes cumulative reward over time.

Unlike supervised learning, which uses training examples with predefined answers, reinforcement learning mainly learns through experience and interaction.

State → Action → Reward → New State

1. Basic Reinforcement Learning Model



2. Agent

An agent is the learner or decision-maker in a reinforcement learning system. It observes the current state, chooses an action, performs the action, receives a reward or penalty, and improves its behavior based on experience.

Example: In a robot-navigation problem, the robot is the agent.

$$A = \{L, R, F, B\}$$

3. Environment

The environment is everything with which the agent interacts. It receives an action from the agent and responds by providing a new state and a reward or penalty. In robot navigation, the room, walls, obstacles, destination, and surrounding objects form the environment.

Agent → Action → Environment
Environment → State + Reward → Agent

4. State

A state represents the current situation of the agent in the environment. It is commonly denoted by S_t . For example, the current position of a robot can represent its state.

$$\begin{aligned} S_0 &= (2, 3) \\ S_{0+1} &= (2, 4) \end{aligned}$$

5. Action

An action is a possible move or decision that the agent can make in a particular state. It is commonly denoted by A_t . The agent selects an action according to its policy.

$$A = \{L, R, F, B\}$$

6. Reward

A reward is the feedback provided by the environment after the agent performs an action. A reward may be positive for a desirable action, negative for an undesirable action, or zero when the action has no significant immediate effect.

$$\begin{aligned} +10 &= \text{Reach destination} \\ -5 &= \text{Hit obstacle} \\ -1 &= \text{Unnecessary movement} \end{aligned}$$

The agent generally tries to maximize the total or cumulative reward rather than only the immediate reward.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

Here, γ is the discount factor that determines the importance of future rewards.

7. Policy

A policy is the strategy used by an agent to decide which action should be taken in a particular state. It is commonly represented by π .

$$\begin{aligned} &\text{Policy: State} \rightarrow \text{Action} \\ &\pi(a, s) \end{aligned}$$

The expression $\pi(a, s)$ represents the policy's probability of selecting action a when the agent is in state s . The purpose of reinforcement learning is to learn a good or optimal policy that produces high cumulative reward.

8. Example: Robot Learning to Reach a Goal

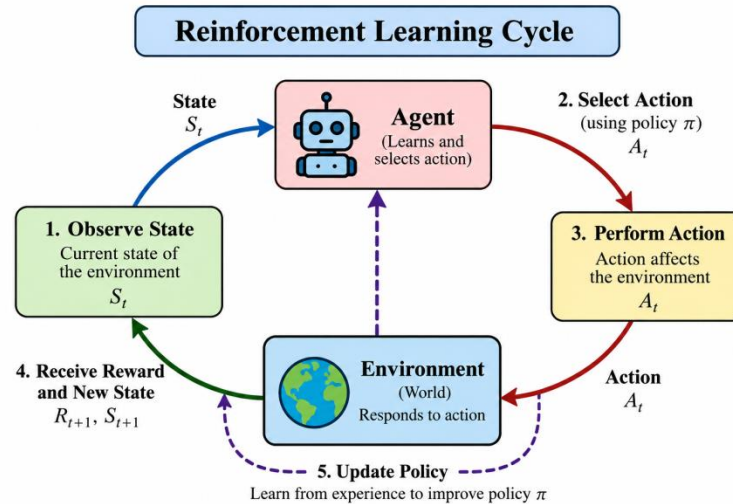
Suppose a robot must learn to reach a goal while avoiding obstacles. Reaching the goal gives a positive reward, while hitting an obstacle gives a negative reward. After repeated interaction, the robot learns which actions lead to better long-term rewards.

$$\text{Robot} \rightarrow \text{Move} \rightarrow \text{Environment} \rightarrow \text{Reward} \rightarrow \text{Learning}$$

Goal reached $\rightarrow +10$

Obstacle hit $\rightarrow -5$

8. Reinforcement Learning Cycle



$$S_t \rightarrow A_t \rightarrow R_{t+1}, S_{t+1}$$

9. Relationship Among the Main Components

Component	Meaning	Example
Agent	Learner or decision-maker	Robot
Environment	World in which the agent operates	Room
State	Current situation	Robot's position
Action	Decision made by the agent	Move right
Reward	Feedback for the action	+10 for reaching goal
Policy	Strategy for selecting actions	Which direction to move

Advantages of Reinforcement Learning

1. The agent learns through experience rather than requiring the correct action for every situation.
2. It is useful for sequential decision-making problems.
3. The agent can adapt its behavior based on rewards and penalties.
4. It can learn strategies for complex environments.
5. It is useful in robotics, game playing, autonomous systems, resource allocation, and optimization.

Observe State $\rightarrow$ *Choose Action* $\rightarrow$ *Receive Reward* $\rightarrow$ *Learn* $\rightarrow$ *Improve Policy*

Q6. Explain Learning from Observations in Artificial Intelligence

Answer: Learning from observations is an approach in Artificial Intelligence in which an intelligent system learns general rules, patterns, or knowledge by observing examples and their outcomes. Instead of explicitly programming every rule, the system analyzes available examples and constructs a general model that can be used to make predictions about new situations.

This type of learning is closely associated with inductive learning. In induction, an AI system moves from specific observations to a general conclusion or hypothesis.

Specific Observations → General Rule

1. Learning from Examples

A common way of learning from observations is to provide the system with a set of input-output examples. An example can be represented as:

$$(x, f(x))$$

Here, x is an input example and $f(x)$ is its observed output. From a collection of examples, the learning system tries to construct a hypothesis h such that:

$$h(x) \approx f(x)$$

The important objective is generalization—the learned hypothesis should produce appropriate outputs for new examples that were not seen during learning.

2. Simple Example

Consider an AI system learning whether a student will pass an examination. It observes examples such as:

Study Hours	Attendance	Result
High	High	Pass
High	Medium	Pass
Low	Low	Fail
Medium	High	Pass
Low	Medium	Fail

From these observations, the system may learn that students with high study hours and good attendance are more likely to pass. It can then use the learned pattern to predict the result for a new student.

3. Main Components

Training Examples: The observations available to the learning system. Each example contains information about a situation and, in supervised learning, its corresponding result.

Target Function: The actual relationship between inputs and outputs that the system is trying to learn.

Hypothesis: The rule or model generated by the learning algorithm from the observations.

Generalization: The ability of the learned model to make correct predictions for new and previously unseen examples.

$$y = f(x)$$
$$h(x) \approx f(x)$$

4. Learning Process

5. Inductive Learning

Learning from observations is often based on inductive reasoning. The system examines specific examples and searches for a hypothesis that explains them.

$$\begin{aligned} & \text{Example}_1 \rightarrow \text{Positive} \\ & \text{Example}_2 \rightarrow \text{Positive} \\ & \text{Example}_3 \rightarrow \text{Negative} \\ & \text{Observed Examples} + \text{Learning Algorithm} \rightarrow \text{Hypothesis} \end{aligned}$$

6. Example: Learning a Classification Rule

Suppose an AI system observes the following examples:

- **Mango 1: Yellow, Sweet → Ripe**
- **Mango 2: Yellow, Sweet → Ripe**
- **Mango 3: Green, Hard → Not Ripe**
- **Mango 4: Green, Sour → Not Ripe**

From these examples, the system may learn the rule:

$$\text{IF Color} = \text{Yellow AND Taste} = \text{Sweet THEN Mango} = \text{Ripe}$$

Thus, the AI system converts individual observations into a general decision rule.

6. Importance in Artificial Intelligence

1. It allows an AI system to improve from experience instead of requiring every rule to be manually programmed.
2. It helps discover useful patterns in observed data.
3. It supports prediction for new and unseen situations.
4. It allows systems to adapt when additional observations become available.
5. It forms the basis of learning techniques such as decision-tree learning and other supervised learning methods.

$$\text{Observations} \rightarrow \text{Learning} \rightarrow \text{Hypothesis} \rightarrow \text{Prediction}$$